

Perceptron y Adaline sobre cada version del dato

Se toman las cinco tablas que dejaron R1 y R2, se parte 70-30 y se entrena el perceptron y el Adaline sobre cada una.

```
clear; close all; clc
pasos =
{'T0_crudo', 'T1_outliers', 'T2_balanceado', 'T3_estandarizado', 'T4_normalizado'};
```

Iris: A) setosa vs resto B) versicolor vs virginica

```
load iris_pasos.mat
vars = {'largo_sepalo', 'ancho_sepalo', 'largo_petal', 'ancho_petal'};
problemas = {'A: setosa vs resto', 'B: versicolor vs virginica'};
R_iris = table();
for pr = 1:2
    for k = 1:5
        T = eval(pasos{k});
        if pr == 1
            X = T(:,vars); y = double(T.clase == 'setosa');
        else
            m = T.clase ~= 'setosa';
            X = T{m,vars}; y = double(T.clase(m) == 'versicolor');
        end
        [acc_p, ep_p, acc_a, ep_a] = entrenar_los_dos(X, y);
        R_iris = [R_iris; table(problemas(pr), pasos(k), ep_p, acc_p, ep_a, acc_a,
        ...
                                'VariableNames',
{'Problema', 'Paso', 'Epocas_perceptron', 'Exact_perceptron', 'Epocas_adaline', 'Exact_adaline'})]];
    end
end
R_iris
```

R_iris = 10×6 table

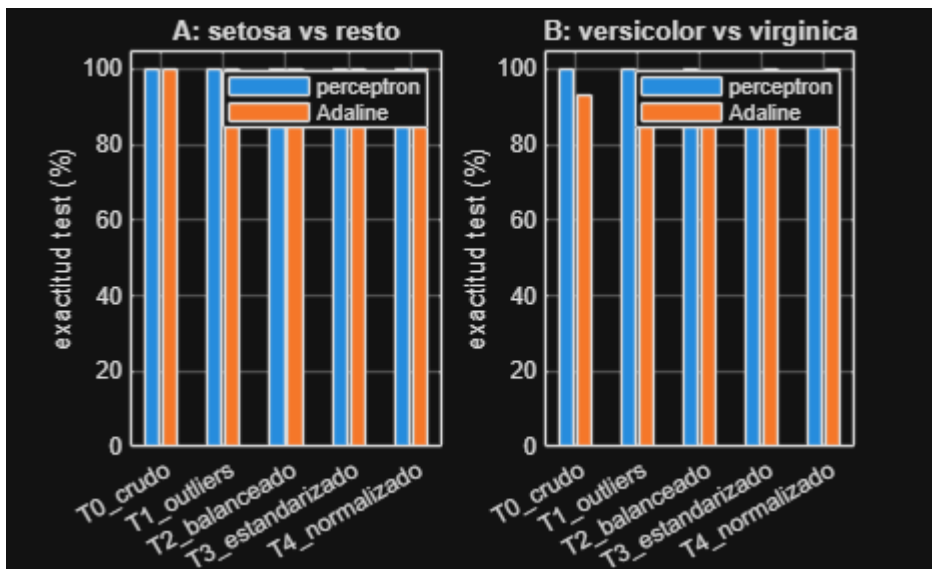
	Problema	Paso	Epocas_perceptron	Exact_perceptron
1	'A: setosa vs resto'	'T0_crudo'	2	100
2	'A: setosa vs resto'	'T1_outliers'	2	100
3	'A: setosa vs resto'	'T2_balanceado'	2	100
4	'A: setosa vs resto'	'T3_estandarizado'	2	100
5	'A: setosa vs resto'	'T4_normalizado'	4	100
6	'B: versicolor vs virginica'	'T0_crudo'	200	100
7	'B: versicolor vs virginica'	'T1_outliers'	200	100
8	'B: versicolor vs virginica'	'T2_balanceado'	200	100

	Problema	Paso	Epocas_perceptron	Exact_perceptron
9	'B: versicolor vs virginica'	'T3_estandarizado'	200	96.6667
10	'B: versicolor vs virginica'	'T4_normalizado'	200	90

```

figure
for pr = 1:2
    subplot(1,2,pr)
    m = strcmp(R_iris.Problema, problemas{pr});
    bar([R_iris.Exact_perceptron(m) R_iris.Exact_adaline(m)]); ylim([0 105]); grid
on
    set(gca, 'XTickLabel', pasos, 'TickLabelInterpreter', 'none')
    ylabel('exactitud test (%)'); title(problemas{pr});
    legend('perceptron','Adaline')
end

```



Banknote

```

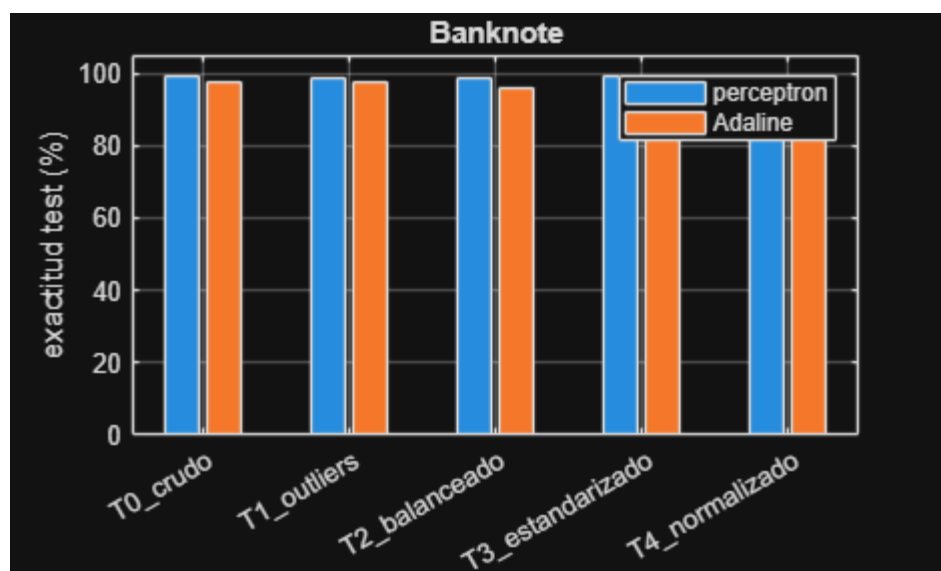
load banknote_pasos.mat
vars = {'varianza','asimetria','curtosis','entropia'};
R_bank = table();
for k = 1:5
    T = eval(pasos{k});
    X = T(:,vars); y = double(T.clase == 'falso');
    [acc_p, ep_p, acc_a, ep_a, ec_a] = entrenar_los_dos(X, y);
    R_bank = [R_bank; table(pasos(k), height(T), ep_p, acc_p, ep_a, ec_a, acc_a, ...
        'VariableNames',
        {'Paso','Filas','Epocas_perceptron','Exact_perceptron','Epocas_adaline','EC_final_adaline','Exact_adaline'})];
end
R_bank

```

R_bank = 5x7 table

	Paso	Filas	Epocas_perceptron	Exact_perceptron	Epocas_adaline
1	'T0_crudo'	1372	200	99.0291	6
2	'T1_outliers'	1372	200	98.7864	6
3	'T2_balanceado'	1220	43	98.9071	6
4	'T3_estandarizado'	1220	23	99.1803	7
5	'T4_normalizado'	1220	42	98.9071	161

```
figure
bar([R_bank.Exact_perceptron R_bank.Exact_adaline]); ylim([0 105]); grid on
set(gca, 'XTickLabel', pasos, 'TickLabelInterpreter', 'none')
ylabel('exactitud test (%)'); title('Banknote'); legend('perceptron','Adaline')
```



Parte 70-30, entrena los dos y devuelve exactitud en test

```
function [acc_p, ep_p, acc_a, ep_a, ec_a] = entrenar_los_dos(X, y)
    rng(2021)
    ind = randperm(numel(y));
    n_tr = round(0.7 * numel(y));
    tr = ind(1:n_tr); te = ind(n_tr+1:end);

    d = 2*y - 1; % perceptron con salida en {-1,1}
    [w, b, ep_p] = perceptron(X(tr,:), d(tr), 3, 0.1, 0, [-1 1], 200, 1);
    acc_p = 100*mean(prededir(X(te,:), w, b, 0, [-1 1]) == d(te));

    [w, b, ep_a, EC] = adaline(X(tr,:), y(tr), 0.01, 1e-3, 500, 1); % Adaline con
    salida en {0,1}
    acc_a = 100*mean(prededir(X(te,:), w, b, 0.5, [0 1]) == y(te));
    ec_a = EC(end);
```

end